Emerging Blockchain Systems:
Solana: Fast Transaction Processing, Minimal Finality Time

Third Generation Blockchains

- By 2025, emerging systems (3rd and 4th generations) address the limitations of earlier generations through innovations like sharding, zero-knowledge proofs, and hybrid consensus mechanisms, enabling faster, greener, and more interconnected networks. Below, I'll explain two representative pairs of pairs (total four pairs)
- This presentation covers Solana – 65,000 transactions per second, and growing
- Use Proof of History + Proof of Stake

Solana

- Solana is a high-performance public blockchain known for its speed, low fees, and ability to scale.
- Launched in 2020, it supports smart contracts, decentralized apps (dApps), and digital assets — much like Ethereum, but faster and cheaper.
- It uses a proof-of-stake (PoS) consensus mechanism alongside a unique feature called proof-of-history (PoH), which timestamps transactions to keep the network in sync. This design helps Solana process thousands of transactions per second with near-instant finality.
- As of June 2025, Solana powers a $198B ecosystem, with its native token, SOL, valued at over $50.15B.
- It ranks sixth globally by market cap and is a favorite among developers aiming to build efficient, scalable blockchain applications.

What is Solana?

- Solana is a public blockchain that supports dApps and crypto projects through smart contracts, which are self-executing programs.
- The Solana ecosystem is supported by its native cryptocurrency, SOL, which allows holders to participate in consensus and governance.
- While it offers features similar to Ethereum, BNB Chain, or Avalanche, Solana prioritizes performance. It is one of the fastest blockchains, focusing on three key aspects: speed, throughput, and cost.
- With Solana's high performance, blockchain-based apps can offer a seamless Web2-like experience. Although Ethereum has dominated Web3, it has often faced congestion issues and high transaction fees, making resource-intensive dApps less practical.
- Solana can, in theory, support up to 65,000 transactions per second (TPS), compared to only around 15 TPS on Ethereum. As per Chainspect, it currently has the highest real-time tips across all chains, overtaking the likes of BNB Chain and Tron.

Speed, Finality and Transaction Cost

- Solana's maximum capacity of 65,000 TPS is no coincidence. It was meant to challenge the throughput of Visa, the world's largest payment provider. Visa has a similar theoretical capacity, and Solana's achievement reflects blockchain technology's potential to compete with traditional finance.
- Besides its high throughput, Solana is also very fast, achieving finality in about 13 seconds. While this is much quicker than Ethereum's 16-minute waiting time, it still lags compared to other high-performance chains like Algorand, Avalanche, and Sonic, which provide near-instant finality. Solana plans to reduce finality by 100 times with the Alpenglow upgrade.
- Solana is also one of the most cost-effective blockchains. The average transaction cost is only $0.01, cheaper than Avalanche or Ton.

The Story

- The story of Solana starts in 2017 with Anatoly Yakovenko, a computer scientist who aimed to improve existing blockchain infrastructure by introducing the PoH concept, which was essential for the network's high performance.
- Yakovenko studied at the University of Illinois Urbana-Champaign and worked at Qualcomm and Dropbox before founding Solana.
- He established Solana Labs in 2017 and released several testnet versions in the following years.
- In 2020, Solana launched its mainnet beta shortly after securing $1.76 million in a public sale of its SOL coin.
- In addition to the public sale, Solana raised hundreds of millions during private token sales.
- It secured over $20 million through several private rounds between 2018 and 2019.
- In June 2021, Solana Labs raised $314.15 million in a Series A round.
- Since then, Solana has been constantly improving its design and adding new features.

How Does it Work?

In what follows, we describe how Solana works (since its launch in 2020 –the situation will drastically change after the upcoming Alpenglow upgrade)

Proof-of-Stake (PoS)

- Solana relies on a PoS version where regular SOL holders delegate their staked coins to validators, who form the network's backbone.
- Validator nodes ensure Solana's functionality and security, creating new blocks with transactions. They earn SOL rewards based on their share of staked coins. Part of the rewards is shared with delegators based on their staking contribution.
- Validators on Solana use high-performance hardware, with requirements much higher than those on Ethereum. For example, the minimum requirements to run a Solana node are 256 GB of RAM (versus 8 GB on Ethereum), a 12-core CPU, and over 1 TB of storage space.
- As of this writing, there are 1,240 validators, 21 of which form the so-called superminority, as they control over 33% of the total stake. Some of the biggest validators include Helius, Binance, Coinbase, Ledger, and SOL Community.
- Solana validators are dispersed across all continents, though with a higher concentration in Europe and North America.

Proof of History

What sets Solana apart is its PoH mechanism, a cryptographic clock that timestamps and verifies the sequence of all network events.

- PoH enables validators to handle transactions in parallel, boosting throughput and reducing congestion. By timestamping each transaction before reaching a block, PoH streamlines the validation process.
- The PoH algorithm is closely associated with Solana's Tower BFT (Byzantine Fault Tolerance) system, a consensus mechanism that helps validators agree even if some nodes fail or behave maliciously.
- Thanks to PoH, validators know the exact order of transactions without having to communicate with each other constantly. This way, Tower BFT reduces the communication requirements typical of other chains.
- Validators vote on the validity of each new block using Tower BFT. The blocks are added if at least 66% of validators approve them.

The Proof of History Process

A Typical Scenario

- **You submit a trade:** A transaction request is sent to the network to exchange Token A for Token B.
- **The transaction enters the PoH sequence:** Before anything else happens, it gets hashed into Solana's ongoing chain of SHA-256 hashes, which assigns it a unique, verifiable timestamp.
- **The timestamp proves order:** Because the hash is part of a sequence, anyone can see when it happened relative to other events, without relying on an external clock.
- **Validators pre-order the transactions:** Thanks to the built-in timeline, validators don't need to waste time figuring out which trade came first. The ordering is already done.
- **Consensus kicks in:** Solana's PoS + Tower BFT handles final agreement and confirmation, but it now works faster because PoH dealt with the timing.
- **Finality and settlement:** Your trade clears in under a second, and you receive Token B. Compared to traditional blockchains, the whole process took fewer steps and less coordination.

Upcoming Solana Upgrades: Firedancer and Alpenglow

- Solana's already impressive performance is about to improve further with the **Firedancer upgrade**, which is about integrating an independent validator client to enhance Solana's infrastructure.
- Developed by Web3 infrastructure firm Jump Crypto, Firedancer could handle over 1M TPS, a massive increase from Solana's current capability. The client software has been running on testnet for years and is set to go live in 2025.
- Probably the biggest upgrade coming to Solana so far is the so-called **Alpenglow**, which will rewrite the network's consensus architecture by eliminating the PoH feature and the Tower BFT, which is used to reach consensus even if some nodes fail.
- The plan for this major overhaul was announced in mid-May 2025, and one of the goals is to reduce transaction finality by 100 times, from about 13 seconds to 100 milliseconds.

Third Generation:

Blockchain	Key Features	How It Differs from Ethereum
Cardano	- Proof-of-Stake (Ouroboros) - Peer-reviewed research - Hydra (Layer 2) for 1M+ TPS - On-chain governance	- **More academic & secure** than Ethereum - Slower development but **no major outages** - **Native multi-asset support** (vs. ERC-20)
Avalanche	- 3 chains: X (exchange), P (platform), C (contract) - Sub-second finality - 4,500+ TPS - Avalanche Warp Messaging (cross-subnet)	- **Faster finality** than Ethereum - **Custom subnets** = private blockchains with shared security - **No gas wars**
Cosmos	- Tendermint BFT consensus - IBC (Inter-Blockchain Communication) - 100+ sovereign chains connected	- **Not a single chain** — a **network of chains** - **Sovereign interoperability** (vs. Ethereum bridges) - Used by Binance Chain, Terra (pre-collapse)